

ANIL MAKHIJA

Senior Mobile Engineer | iOS (Swift/SwiftUI) • React Native • BLE & IoT Connected Devices • Architecture & Security

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PROFESSIONAL SUMMARY

Senior Mobile Engineer with 11+ years of experience building and leading native iOS (Swift, SwiftUI) and cross-platform (React Native, TypeScript) applications — with deep, production-proven expertise in BLE (CoreBluetooth), WiFi, and IoT connected-device ecosystems that consistently differentiates my work from typical mobile engineers. Built and shipped multiple consumer and professional-facing apps that pair with, configure, and monitor physical IoT devices in real time — solar energy inverters, batteries, and smart hardware — serving 50,000+ users with reliable multi-device discovery, real-time telemetry, and OTA firmware update workflows. Expert in modern iOS architecture (MVVM, VIPER, Clean Architecture, SPM modularisation) and Swift Concurrency (async/await, Actors), with 5+ years of React Native delivering iOS + Android from shared TypeScript codebases. Production-hardened mobile security: Keychain, biometric authentication, OAuth 2.0/JWT, certificate pinning, and AES-256 encryption — applied across IoT, HealthTech, and financial-benefit platforms handling sensitive data. Proven technical leader: led 6-member teams, mentored engineers, drove architecture decisions, and delivered measurable outcomes (20% performance gains, 30% fewer connectivity incidents, 25% engagement growth) across distributed Agile teams.

BLE & IoT CONNECTED DEVICE EXPERTISE

Production CoreBluetooth & Multi-Device Systems

- Full CoreBluetooth stack: CBCentralManager setup and background state restoration, CBPeripheral discovery, GATT service/characteristic enumeration, read/write/notify operations, connection state management, MTU negotiation for throughput optimisation — production-shipped across multiple consumer and installer applications.
- Designed concurrent multi-device BLE management (up to 5 simultaneous peripheral connections) using Swift Concurrency (Actors, TaskGroup) — Actor-isolated device state prevents data races across simultaneous connections, verified with Thread Sanitiser in production builds.
- Built robust reconnection strategies: exponential backoff, connection timeout handling, out-of-range detection, and background BLE state restoration — ensuring reliable device pairing even after app termination or backgrounding.
- Designed real-time BLE characteristic notification pipelines using Swift Concurrency AsyncSequence — transforming raw peripheral callbacks into typed, structured device-state events consumed reactively by SwiftUI views.

WiFi Provisioning, OTA Updates & Firmware Collaboration

- Implemented WiFi device provisioning: delivering network credentials to IoT devices via BLE characteristic write, guiding users through SSID selection and provisioning confirmation with real-time status feedback.
- Built OTA firmware update workflows: cloud firmware download with checksum validation, chunked binary transfer over BLE, progress monitoring, and retry/rollback handling for failed updates — a low-friction, reliable field update experience.
- Collaborated directly with embedded/firmware engineering teams to define GATT profile specifications — service UUIDs, characteristic properties, binary protocol encoding — bridging mobile app development with hardware communication contracts.

Real-Time Data Pipelines & Cloud-Connected IoT Architecture

- Designed end-to-end event-driven data pipelines: BLE/WiFi device telemetry streamed through WebSocket/MQTT into cloud backend services, surfaced through REST/GraphQL APIs to mobile clients — full edge-to-cloud-to-client architecture.
- Built offline-first, reliable device state synchronisation (Core Data with background sync and conflict resolution) — ensuring data integrity for connected-device apps operating in real-world, often degraded-connectivity environments.
- Applied algorithmic optimisation to real-time telemetry rendering: resolved $O(n^2)$ chart rendering bottlenecks through indexed/sorted data structures, sustaining 60fps on live, high-frequency IoT data visualisations.

CORE TECHNICAL SKILLS

- **iOS Platform:** Swift (expert), SwiftUI, UIKit, Combine, Objective-C, Core Data, Core Animation, Core ML, Xcode, Instruments

- **React Native & Cross-Platform:** React Native (TypeScript/JavaScript, 5+ years) — cross-platform iOS + Android; Redux Toolkit, RTK Query, Context API; Native Module bridge development (Swift/Kotlin)
- **BLE & IoT (Core Strength):** BLE (CoreBluetooth), WiFi provisioning, OTA firmware updates, WebSocket, MQTT — connected device communication (primary differentiator)
- **Architecture:** MVVM, VIPER, Clean Architecture, Coordinator Pattern, SOLID, Protocol-Oriented Programming, SPM Modular Architecture / Micro Frontend decomposition
- **Concurrency:** Swift Concurrency (async/await, Actors, Structured Concurrency, AsyncSequence), GCD, OperationQueue, Thread Safety
- **Security:** Keychain, Biometric Auth (Face ID/Touch ID), Certificate Pinning, CryptoKit (AES-256), OAuth 2.0/JWT, App Transport Security
- **APIs & Networking:** RESTful APIs, GraphQL, WebSocket, MQTT, JSON (Codable), API contract design with backend teams
- **Testing:** XCTest, XCUITest, Quick/Nimble, Jest, React Native Testing Library, Detox, TDD — unit, integration, UI, and E2E testing
- **CI/CD & Release:** Fastlane, GitHub Actions, Git (GitFlow), CI/CD pipelines, App Store Connect, TestFlight, Google Play Console
- **Observability & Cloud:** Firebase (Analytics, Crashlytics, Performance Monitoring, Remote Config feature flags/A-B testing), AWS (basic)
- **AI-Assisted Development:** Claude AI, GitHub Copilot — AI-assisted development for refactoring, test generation, and documentation
- **Leadership & Process:** Agile/Scrum, technical leadership, mentoring, code reviews, cross-functional collaboration (Product, Design, Backend, QA), distributed team communication (India-US)

WORK EXPERIENCE

Senior Software Engineer – iOS & IoT | SMA Solar Technology AG

Jul 2025 – Dec 2025

SMA Energy App (Consumer) | SMA 360° App (Installer) | Bengaluru, India

Swift · SwiftUI · Combine · BLE · WiFi · OTA Updates · MVVM · REST APIs · CI/CD · App Store

- Built two production iOS apps integrating real-time BLE/WiFi IoT device communication — multi-device discovery (up to 5 inverters), OTA firmware updates, and system configuration — reducing installer commissioning time ~40%.
- Applied Swift Concurrency (async/await, Actors, TaskGroup) for concurrent multi-device BLE discovery — eliminated threading bugs and improved responsiveness across simultaneous device sessions.
- Collaborated with firmware/embedded teams to define GATT communication protocol; managed full App Store release lifecycle with Fastlane CI/CD automation.

Lead iOS Consultant – IoT & Mobile Architecture | Encora Innovation Lab Pvt. Ltd. (Client: Enphase Energy)

Aug 2022 – Jun 2025

Enphase Enlighten — Cloud-connected IoT energy platform for 50,000+ users | Bengaluru, India

Swift · SwiftUI · React Native · TypeScript · BLE · WiFi · MQTT · MVVM · Clean Architecture · SPM Modules · Security · TDD · CI/CD · Mentoring

- Led a 6-member team building Enphase's connected smart energy platform — BLE/WiFi multi-device management, real-time IoT telemetry via WebSocket/MQTT, OTA firmware delivery, and offline-first device state sync for 50,000+ users.
- Designed Actor-isolated BLE device manager with concurrent multi-peripheral connection handling and AsyncSequence-based real-time telemetry pipeline — zero data races verified via Thread Sanitiser.
- Decomposed monolithic codebase into 6 SPM feature modules with protocol-based API contracts, enabling parallel team delivery; applied MVVM/Clean Architecture with TDD (XCTest, XCUITest) as CI-enforced practice.
- Built React Native (TypeScript) components alongside native iOS — Native Module bridges exposing BLE and biometric capabilities to the JavaScript layer for cross-platform feature parity.
- Implemented production-grade security: Keychain, biometric auth, OAuth2/JWT, certificate pinning, AES-256 encryption — protecting sensitive customer energy and account data.
- Improved app performance 20%, reduced connectivity incidents 30%, increased user engagement 25%; mentored 6 engineers through code reviews and architecture coaching.

Software Engineer – iOS & React Native | Legato Health Technologies

Aug 2020 – Jul 2022

Sydney Health — Consumer health & financial benefits app, 10,000+ users | Bengaluru, India

Swift · SwiftUI · React Native · TypeScript · MVVM · Security · REST APIs · TDD · Agile

- Built Sydney Health (React Native + SwiftUI/MVVM) — insurance benefit management, claims, and identity verification for 10,000+ patients, with full biometric/Keychain/OAuth2 security implementation.
- Delivered cross-platform iOS + Android features from shared TypeScript codebase; drove 15% engagement increase through data-driven UX improvements.

Senior Software Engineer – iOS | UST (Client: Legato Health Technologies) Jan 2020 – Aug 2020

Bengaluru, India

Swift · SwiftUI · MVVM · TDD · XCTest · Agile

- Delivered 3 production iOS features using MVVM and TDD — 100% on time, 2 weeks ahead of schedule; collaborated cross-functionally under Agile constraints.

Software Engineer – iOS & Cordova | Synoverge Technologies Pvt. Ltd. Dec 2016 – Nov 2019

VFS Global, Roquette App | Ahmedabad, India

Swift · Objective-C · Cordova · JavaScript · MVVM · MVC · REST APIs · App Store · Play Store

- Shipped 10+ cross-platform mobile apps (iOS + Android via Cordova) using Swift, Objective-C, and JavaScript; 3 apps live on App Store with 30% faster load times through performance optimisation.

Sr. iOS Developer | iOS App Developer | AIS Technolabs / Techseria Aug 2014 – Nov 2016

Ahmedabad, India

Swift · Objective-C · MVC · CoreML · REST APIs · CI/CD

- Built 13+ iOS apps integrating CoreML, Unity, and REST APIs; established core iOS engineering foundations supporting 11 years of continued technical growth.

EDUCATION

Master of Computer Applications (M.C.A.) — Hemchandracharya North Gujarat University | Aug 2011 – May 2014

Focus Areas: Programming Languages, Mobile Development, Distributed Systems, Software Architecture

HONOURS & RECOGNITION

- Encora Inc – Extraordinary Client Service (2025): Recognised for BLE/IoT engineering excellence and connected-device platform delivery at scale.
- Encora Inc – Living Our Values (Mar 2023): Recognised for technical leadership, mentoring, and cross-platform engineering excellence.
- Legato Health Technologies – GoAbove Impact Award (Mar 2022), Team Player (May 2022), Individual Contributor (Apr 2021).